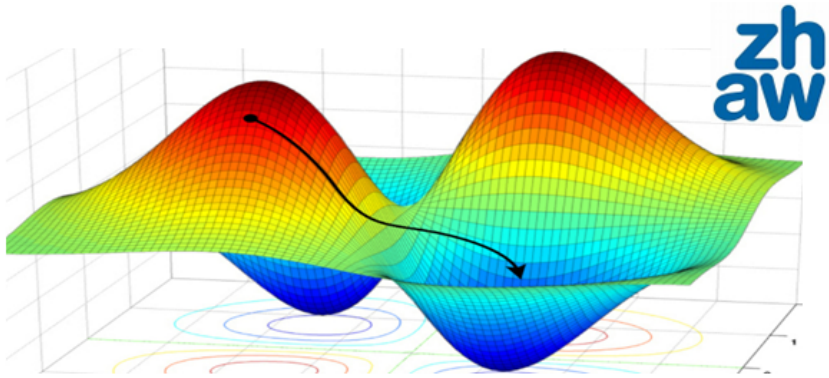

Projects

As part of the module examination, you must carry out a project in teams of three or four people. The deliverable consists of a report and a short pitch.

Tasks

1. Choose/find an optimisation problem (list of ideas see below)
2. Describe the problem and formulate it in mathematical terms
3. Implement at least two optimisation methods to solve the problem
4. Compare the results in terms of accuracy/effectiveness, efficiency and robustness
5. Discuss the results, what are your conclusions?
6. Summarise your project in a short report (pdf format), max. 4 pages without appendix
7. Add the code (and possibly additional figures) as well as references in the appendix of the report
8. Present your results in a short pitch on the last day of the module (max 5min). Focus on the main points.
9. The final reports can be uploaded within 2 weeks after the module

Evaluation Criteria

You can achieve a maximum of 15 points. These are the assessment criteria for the above tasks.

1. Choice of optimisation problem: complexity of problem, novelty of the problem (self-invented or just copied 1:1 from a uncited source?): 0-4 points (plagiarism leads to failure of the module, penalty up to -15 points)
2. comprehensible, concise, correct?: 0-2 points
3. Methods appropriate? Correctly implemented?: 0-2 points
4. Results and comparison correct?: 0-3 points
5. Conclusion comprehensible, concise, correct?: 0-1 points
6. Report fulfills requirement?: penalty up to -3 points
7. References clear, code runable? if not, penalty up to -3 points
8. Quality of presentation/pitch (clarity, correctness, conciseness, answers to questions): 0-3 points

Remarks

1. The time must be strictly adhered to
2. Please register your group in Moodle

Support

There will be special time slots in which you can discuss your project with the lecturers (see Moodle).

Choice of topic

Choose an interesting optimization problem and try and compare different - and potentially novel - algorithms. Do not look for the easiest ways (easy problem, standard algorithms). Invest in at least one thing: the formulation of an interesting problem, a novel algorithm, or an in-depth comparative study.

Some ideas:

1. Choose a relatively simple problem (e.g., a low-dimensional constrained continuous problem) and examine the performance of different algorithms in detail
2. Get inspiration from the exercises and find a similar (typically more demanding) real world problem
3. Choose a new type of problem, e.g., traffic flow optimization problems, DNA sequence alignment problems, protein structure predictions...
4. Study strategies for hyperparameter tuning of a machine learning model with different optimization methods
5. Use ant optimization or another swarm optimization algorithms for a relatively simple problem

We can also give you more specific ideas, if you are really stuck.